

Molecule polarity

Learning outcomes

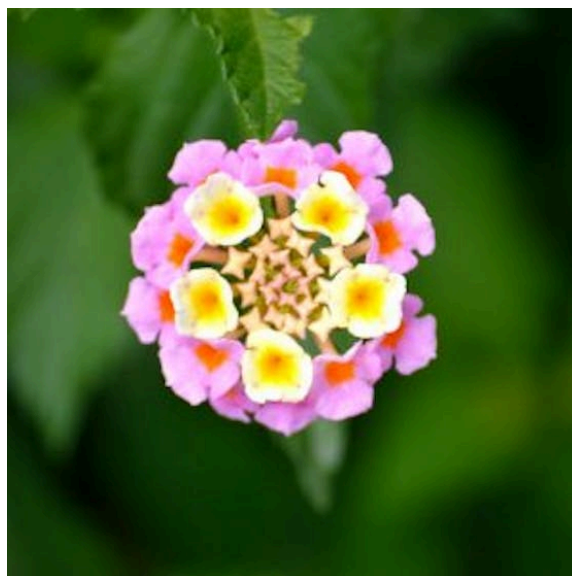
By the end of this section you should be able to deduce the net dipole moment of a molecule or ion by considering bond polarity and molecular geometry.

Symmetry is an important part of our everyday life and the world around us. It is a term used to describe an identical relationship between two or more parts of an object. If you were to cut a symmetrical object in half, the two halves should look the same. Just pick up a mirror and look at your face to see symmetry in action!

Symmetry allows us to create patterns that we then use to organise the world around us. Many of the animals and plants around us demonstrate symmetry and we attribute their beauty to it, including the wings of a butterfly or in a flower such as the *Lantana camara*, native to central and south America, which can be seen in **Figure 1**.



Credit: Giantstep Inc, Getty Images



Credit: Victoria1988, Getty Images

Figure 1. The wings of a *Morpho enelaus* butterfly and the flower of the *Lantana camara* demonstrate symmetry.

Symmetry may also be displayed on a molecular level and can affect the properties of a molecule.

Molecular polarity

Recall your understanding of drawing Lewis formulas from [section S2.2.2](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/multiple-covalent-bonds-id-45128/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/multiple-covalent-bonds-id-45128/>). Use this knowledge to draw a molecule of CO_2 , and link this to what you learned about electronegativity and polar bonds in the last section. Use [section 9](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/first-ionisation-energy-electron-affinity-and-id-45111/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/first-ionisation-energy-electron-affinity-and-id-45111/>) in the DP chemistry data booklet to help you determine whether the bonds within carbon dioxide are polar.

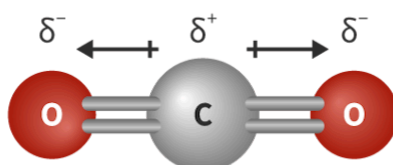


Figure 2. Carbon dioxide has polar bonds. More information for figure 2

The diagram illustrates a carbon dioxide (CO₂) molecule. In the center, there is a carbon atom (C) connected to two oxygen atoms (O) on either side. The carbon atom is shown in gray, and the oxygen atoms are in red. Each bond between the carbon and oxygen atoms is represented by two parallel lines, indicating double bonds. Above these double bonds, there are symbols showing the polarity: $\delta+$ above the carbon atom and $\delta-$ above each oxygen atom. Arrows point from the carbon atom towards each oxygen atom, indicating the direction of polarity. The diagram labels the oxygen atoms with a partial negative charge ($\delta-$) and the carbon atom with a partial positive charge ($\delta+$). This highlights the polar nature of the bonds due to the electronegativity difference between carbon and oxygen.

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The carbon dioxide molecule has polar bonds (**Figure 2**). Based on your knowledge of electrostatic attraction and the fact that opposite charges attract, you may expect that one CO₂ molecule would have an attraction for another CO₂ molecule as their opposing partial charges should attract each other.

Consider the molecular geometry of CO₂ using your knowledge from [section S2.2.4 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/vsepr-id-45129/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/vsepr-id-45129/). The central carbon atom has 2 electron domains and no non-bonding pairs, resulting in a linear shape and a bond angle of 180° (**Figure 3**).

Although there are polar bonds present, the molecule has a **symmetry** that prevents the molecule as a whole from having a net 'positive end' (positive dipole) and a net 'negative end' (negative dipole). How do you think this would affect the ability of one CO₂ molecule to attract another CO₂ molecule?

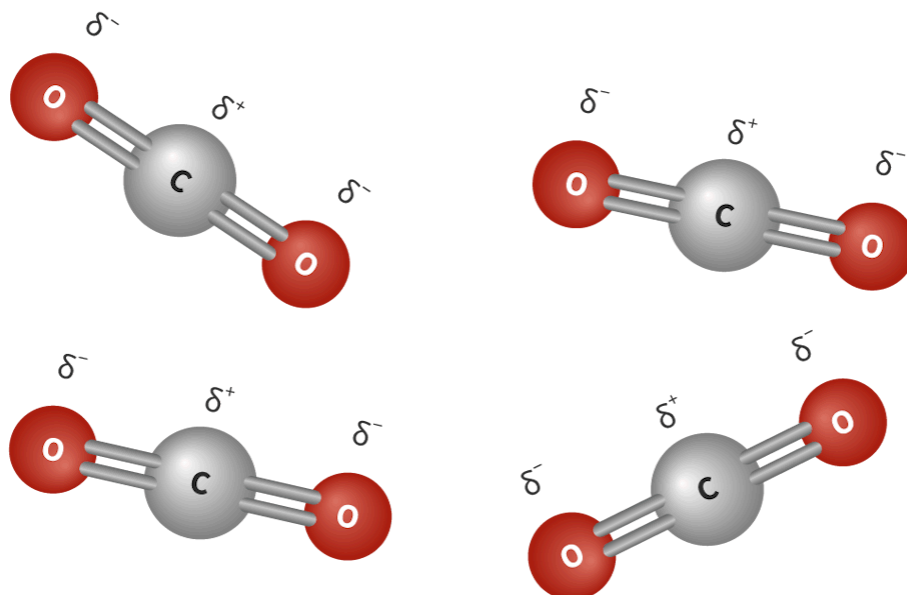


Figure 3. Carbon dioxide has polar bonds.

 More information for figure 3

The image depicts four carbon dioxide (CO₂) molecules, each with a central gray carbon atom bonded to two red oxygen atoms. The bonds between carbon and oxygen are shown as double lines, indicating double bonds. The image includes symbols for partial charges (denoted as δ^+ and δ^-) near the carbon and oxygen atoms, respectively, showing the polarity of the bonds. The carbon atom carries a partial positive charge (δ^+) while the oxygen atoms carry partial negative charges (δ^-). This illustrates the concept that although CO₂ has polar bonds due to the difference in electronegativity between carbon and oxygen, the molecule itself is linear and symmetric, resulting in no net dipole moment overall. This symmetry prevents the molecule from having distinct positive and negative ends, affecting its interaction with other molecules.

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Compare this to a molecule of hydrogen chloride, HCl, which has a polar bond between hydrogen and chlorine. Although the molecule is linear, in contrast to CO₂, HCl has a clear net positive end (positive dipole) and a net negative end (negative dipole). What do you think happens when many molecules of HCl are together?

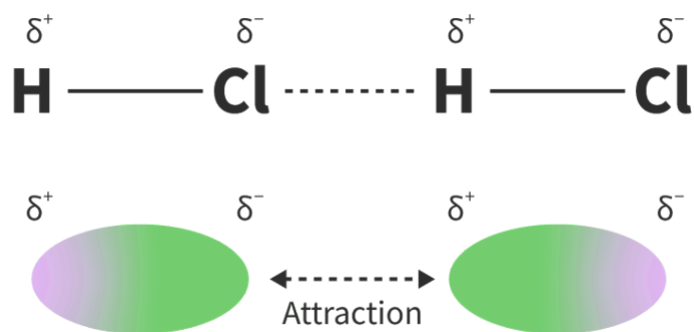


Figure 4. Hydrogen chloride has polar bonds and is a polar molecule.

 More information for figure 4

The diagram illustrates the polar nature of hydrogen chloride (HCl) molecules. At the top, two HCl molecules are shown, with positive dipoles (δ^+) at the hydrogen atom and negative dipoles (δ^-) at the chlorine atom. The bonds between the atoms are depicted, with an arrow indicating their polar nature. Below, two overlapping ellipses represent the molecules, illustrating the attraction between opposite dipoles (δ^+ on hydrogen and δ^- on chlorine). The depiction emphasizes the molecular attraction and alignment due to the polar character of HCl.

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The presence of a net dipole on the molecule means that the positive dipole of one molecule is attracted to the negative dipole of another molecule. In the case of HCl, there are polar bonds **and** the overall molecule is polar (**Figure 4**). This allows the molecules to orient with opposite partial charges aligned and to pack closely together, in a way a non-polar molecule like CO_2 cannot do.

Summary

We can draw an important conclusion from this: it is the combination of both polar bonds **and** symmetry that determines this important property for the overall molecule.

An asymmetrical molecule with polar bonds will have an overall net dipole moment, i.e. a positive end and negative end (**Figure 5**).

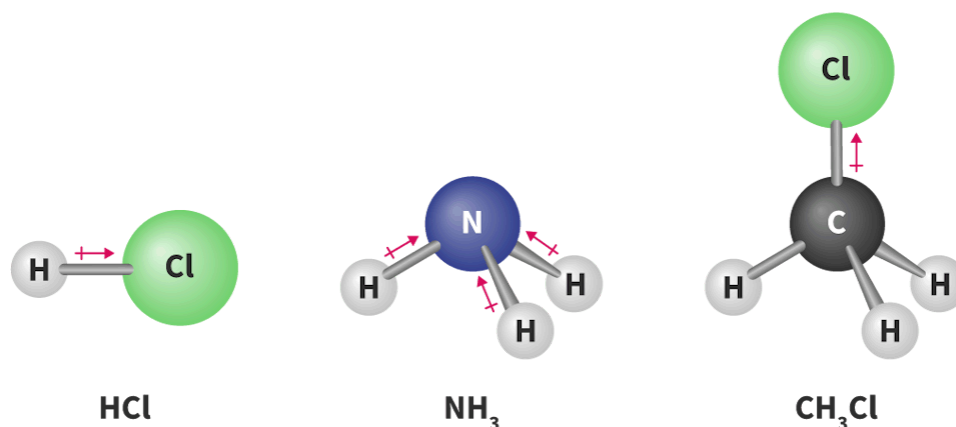


Figure 5. Molecules with bond dipoles and a net dipole moment.

 More information for figure 5

The image shows three molecular structures representing molecules with bond dipoles and a net dipole moment. The first molecule is HCl (hydrogen chloride), where a hydrogen atom is bonded to a chlorine atom. An arrow points from the hydrogen to the chlorine, indicating the dipole direction with a '+' near hydrogen showing it's the positive end.

The second molecule is NH₃ (ammonia), consisting of a nitrogen atom bonded to three hydrogen atoms arranged in a trigonal pyramidal shape. Arrows point from each hydrogen to the nitrogen, showing bond dipoles converging at the nitrogen.

The third molecule is CH₃Cl (chloromethane), where a central carbon atom is bonded to three hydrogen atoms and a chlorine atom. An arrow points from the central carbon to the chlorine atom, indicating the net dipole direction with a '+' near carbon.

Overall, the image illustrates how bond dipoles lead to an overall net dipole moment in these molecules, influencing their polarity.

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A molecule may have polar bonds present but the symmetry of the molecule may prevent the molecule as a whole being polar if the dipole moments of the individual bonds cancel out (**Figure 6**).

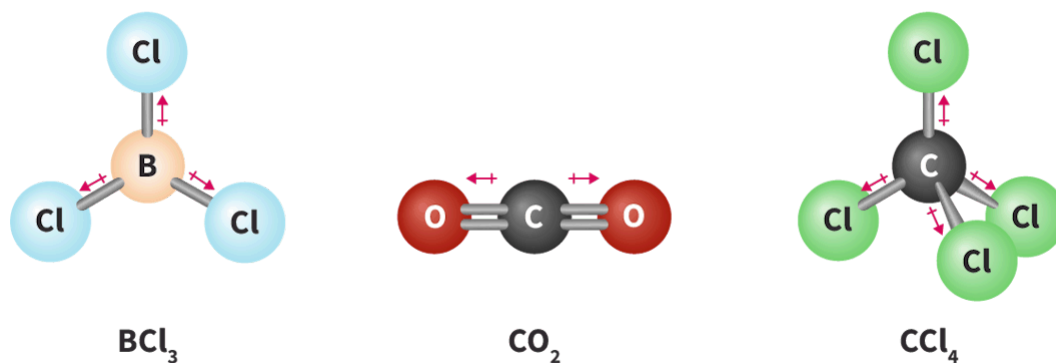


Figure 6. Non-polar molecules: no net dipole moment because the symmetry of the molecule causes the dipoles to cancel out.

 More information for figure 6

The image displays three molecular structures: BCl_3 , CO_2 , and CCl_4 , each with arrows indicating the direction of dipole moments. BCl_3 is structured with a central boron atom surrounded symmetrically by three chlorine atoms. The arrows show equal but opposite dipole directions canceling each other out, resulting in no net dipole moment. CO_2 is a linear molecule with a central carbon atom double-bonded to two oxygen atoms, and arrows pointing outwards from the carbon indicate dipole moments cancel each other due to the linear configuration. CCl_4 shows a central carbon atom tetrahedrally bonded to four chlorine atoms; the dipoles are directed symmetrically from the center, resulting in a net dipole of zero. These structures illustrate the concept that molecular symmetry can lead to the cancellation of dipole moments, making the overall molecule non-polar.

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Study skills

The terms bond polarity and molecular polarity can be confused. Remember that just because polar bonds are present in a molecule, it does not mean the entire molecule is polar.

If a molecule has polar bonds but is non-polar overall then it is because the symmetry of the molecule causes the dipoles to cancel out. You must first identify the molecular geometry and determine if the molecule shows symmetry. Refer to symmetry in any answers you give — this is a key term.

In this activity, you will link together your knowledge to draw Lewis diagrams, use VSEPR theory to determine the shape of the molecule, identify whether polar bonds are present and determine whether the overall molecule is polar based on its symmetry (which you have learned in this section).

Activity

- **IB learner profile attribute:** Inquirers
- **Approaches to learning:**
 - Social skills — Working collaboratively to achieve a common goal
- **Time required to complete activity:** 30 minutes
- **Activity type:** Pair activity

Overall molecular polarity is based on the presence of polar bonds. To determine whether a molecule is polar, the Lewis formula of the molecule must be drawn and the shape determined using VSEPR theory. Electronegativity values can then be applied to each bond within the molecule.

This task will allow you to explore these concepts.

Before you begin:

Using what you have learned about Lewis formula, polar bonds and molecular polarity:

1. Draw the Lewis formula for each of the molecules below .
2. Determine the molecular shape using VSEPR.
3. Annotate your diagram to indicate if polar bonds are present.
4. Conclude if the molecule has an overall dipole moment.
5. Check your answers by swapping your diagrams with your partner. Discuss any differences between your results.

Molecules:

- Hydrogen fluoride, HF
- Chloroform CHCl_3
- Ammonia, NH_3
- Hydrogen cyanide, HCN
- Borane, BH_3
- Fluoromethane, CH_3F

- Tetrafluoromethane, CF_4